unfound

PREPARED FOR TIDE · OCTOBER 2026

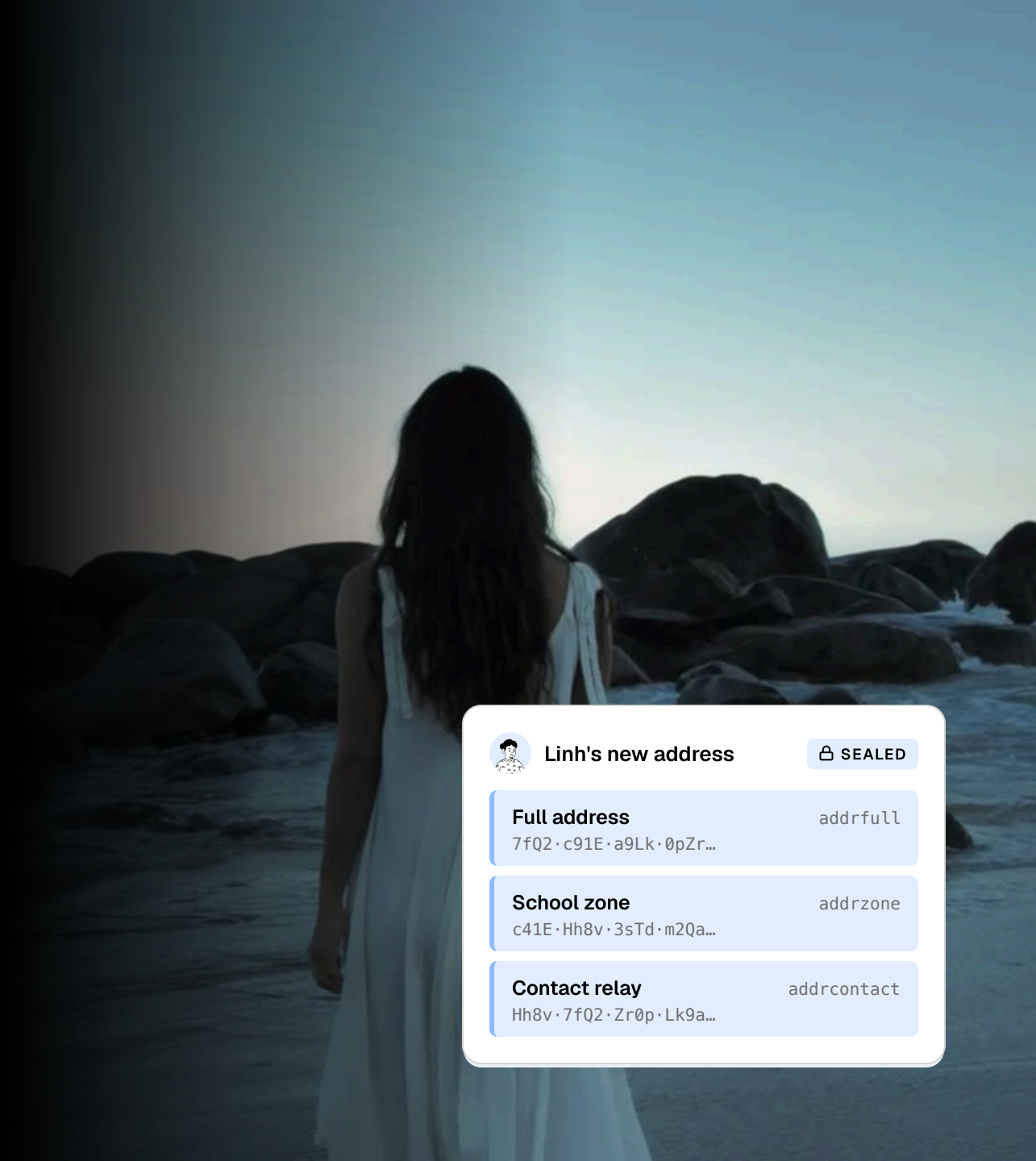
Be reached, not found.

Every school. Every clinic. No one looks you up.

A safe-address layer for family-violence survivors. Services can reach them after they move. No employee, admin or operator can look them up.



Seven people. One address.
No master key.



Leaving is when the risk peaks. The new address is what the abuser looks for.

AU · PREVIOUS PARTNER

1.5M

Australian women have experienced violence from a **previous** partner, nine times more than from a current one. [1]

AU · MOVED OUT

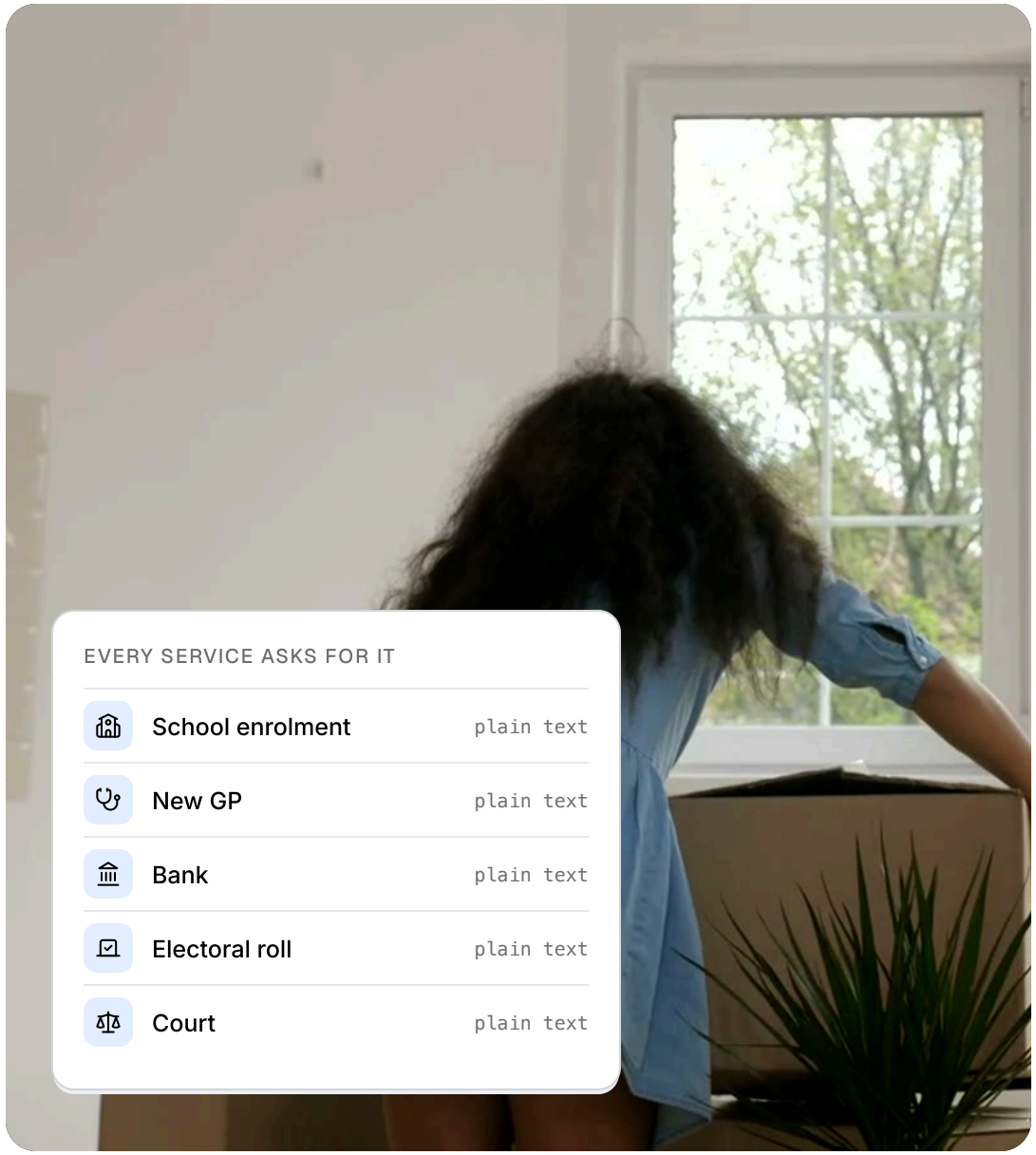
866,700

of them moved out when it ended. Each move is a new address in school, GP, bank and court records. [1]

NSW · SEPARATION

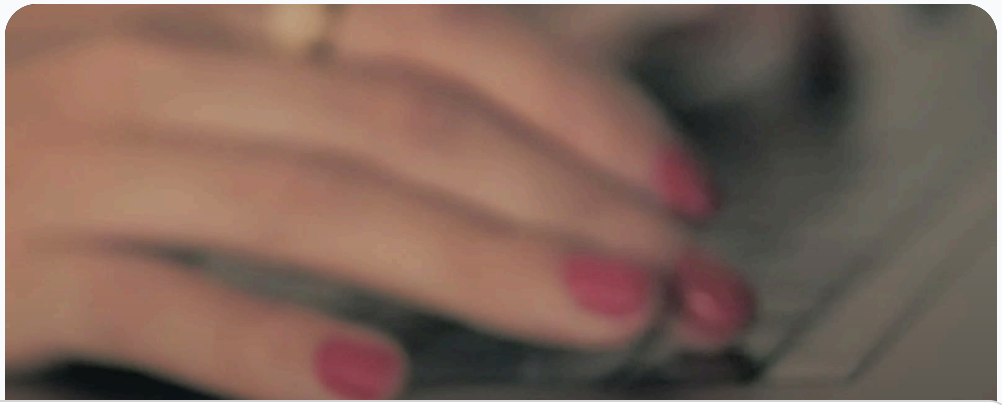
2 in 3

women killed by an intimate partner were killed after leaving or while separating. [2]



[1] ABS, Personal Safety Survey 2021–22 (accessed 22 Oct 2026). [2] NSW DV Death Review Team 2021–23, via Canberra Times (accessed 26 Oct 2026). Photo: Mixkit.

Not a hacker. One person with legitimate access, doing one lookup.



QLD · 7 AGENCIES · 4 YEARS

2,215

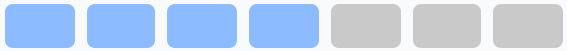
misuse-of-information allegations. Outside police: **4.8×**. [3]



UK REGULATOR · 2022–23

4 of 7

organisations reprimanded over victims' data had **sent a safe address to the abuser**. [4]



Records · admin view

Dan K. · IT admin · search: "Linh"

LEGACY SYSTEM

Nguyen, T.	14 Albert St, Footscray VIC 3011	
Linh P.	Unit 4, 18 Moreland St, Footscray VIC 3011	VIEWED
Okafor, J.	2/91 Barkly St, Footscray VIC 3011	
Smith, R.	7 Hyde St, Yarraville VIC 3013	

No approval. No alert. The survivor never knows.

2014 · QUEENSLAND

An officer searched the police database and gave a survivor's new address to her abuser. [5]

2023 · ENGLAND

A council posted a letter with a survivor's new address to her old home. Her ex-partner opened it. [6]

[3] CCC Qld, Operation Impala Ex. 95. [4] UK ICO, 29 Sep 2023. [5] QCAT via Sydney Criminal Lawyers. [6] ICO reprimand, Charnwood BC, 29 Nov 2023. Accessed 22–27 Oct 2026. Records are synthetic.

Every fix today still trusts someone who holds the key.

STILL TRUSTS A PERSON

Role-based access + audit logs

Admins configure and override them. Logs show harm after it has happened.

Still readable by: every admin

STILL TRUSTS A PERSON

Encryption at rest

The provider holds the key, so the operator, and anyone who breaches them, reads everything.

Still readable by: the operator

STILL TRUSTS A PERSON

Silent elector (AU)

Hides the address on the electoral roll only. Schools, clinics and courts still store it.

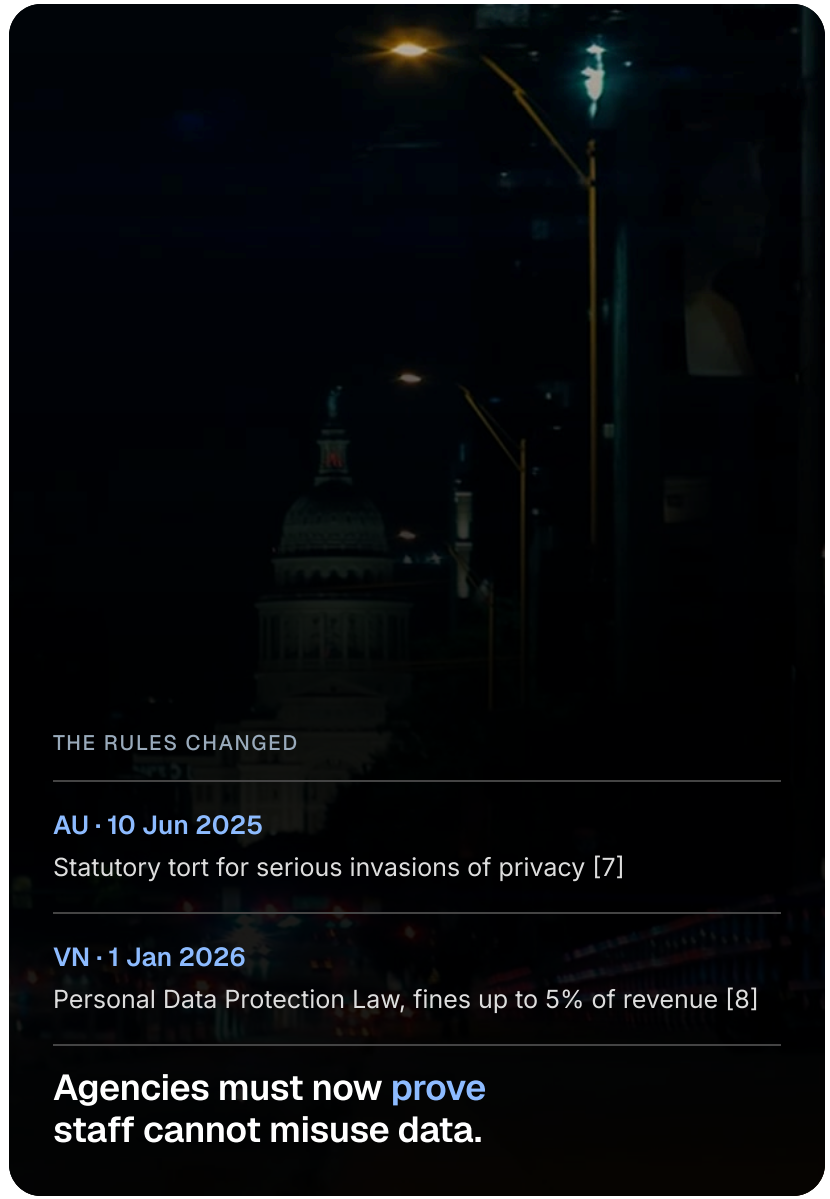
Still readable by: every other service

STILL TRUSTS A PERSON

Address Confidentiality Programs (US, ~39)

Manual mail forwarding. No cryptographic control, and nothing for the survivor to see.

Still readable by: program staff



THE RULES CHANGED

AU · 10 Jun 2025
Statutory tort for serious invasions of privacy [7]

VN · 1 Jan 2026
Personal Data Protection Law, fines up to 5% of revenue [8]

Agencies must now prove staff cannot misuse data.

[7] Lander & Rogers. [8] Tilleke & Gibbins; Conventus Law. ACP count: NACAP to NASS (2022). Accessed 26 Oct 2026. Photo: Mixkit.

One address. Three sealed parts.
Each service opens only its own.



Westgate Primary ZONE ✓
In Footscray North zone



GP clinic RELAY
Suburb and a relay number



Court mail AT PRINT
Street opens as the label prints



Linh · survivor SEALED
Sealed once, on her phone

Full address	addrfull
quorum only · one view	
School zone	addrzone
catchment only	
Contact relay	addrcontact
suburb + relay number	

Who looked: every attempt, approval and view



Const. Reyes 2 OF 3
One view, then revoked



Dan · IT admin 412
Ciphertext; self-grant pending



Unfound · operator NO KEY
Stores ciphertext only

SCOPE

What Unfound protects, and what it doesn't.

One part of a safety plan. It closes the path survivors cannot close themselves: the services that hold their address.

40% of UK survivors had their new address disclosed to the abuser, **mostly by agencies such as their child's school or the council**. **52%** for survivors with children. [1]



OUR ESTIMATE — Spares about **1 in 12** survivors that disclosure in a pilot, up to **1 in 3** if every service joins.

[1] Citizens Advice / Opinium survey of 1,019 survivors, fieldwork Sep–Oct 2019, published 28 Feb 2020. Estimate method: brief §6.3. Accessed 27 Oct 2026.

VERIFIED ON LIVE TIDE	Licensed realm with QEA; enrolment blocked by ORK availability, so no live decrypt yet.
IMPLEMENTED IN CODE	Sign-in with DPoP, doEncrypt / doDecrypt per tag, AdminAPI change sets.
SIMULATED IN THE PROTOTYPE	The eight-scene story on a mock fabric, relay and court printing.

INSIDE THE BOUNDARY · SERVICES THAT JOIN — CLOSED

- **Staff or a hijacked account looks her up**
  Opens only its own tag; the attempt is logged ✓
- **An admin grants themselves access**
  Change set, 2 of 3 approvers, or 412 ✓
- **Her son's school, clinic or pickup record**
  Same sealed parts: not a back door to the household ✓
- **A letter or court form carries the street**
  Filled in only at print, never saved ✓
- **Unfound's servers are breached**
  Ciphertext only, no key beside it ✓

OUTSIDE THE BOUNDARY — NEEDS OTHER SUPPORT

- Following, GPS trackers, shared location
- A service that has not joined keeps a copy
- Her own phone, if taken (reduced, not prevented)
- Friends or family who already know
- Notes taken after a legitimate view

Safety planning, tech-safety checks, protection orders

The promise only holds if no one can hold the key, or grant themselves access.



Survivor's browser

HERMETIC E2EE

```
tc.doEncrypt([
 { data: street, tags: ["addrfull"] },
 { data: zone, tags: ["addrzone"] },
 { data: relay, tags: ["addrcontact"] }
])
```

Sealed before it leaves the phone. Sign-in with BYOiD; tokens bound by **DPoP**.

Tide Fabric

INDEPENDENT ORK NODES

14 OF 20

Keys exist only as fragments. Before helping decrypt, the ORKs check the caller's tag role: `_tide_addrzone.selfdecrypt` . The network refuses, not our UI.

Unfound store

CIPHERTEXT ONLY

```
AAFb7fQ2c91Ea9Lk0pZrHh8v3sTdm2QaZr0pLk9aXc4E…
AAGc41EHh8v3sTdm2Qa7fQ2c91Ea9Lk0pZrB1x…
AAHh8v7fQ2Zr0pLk9a3sTdc41Em2QaXw8…
```

No key on our servers to steal, leak or subpoena.

QUORUM-ENFORCED GOVERNANCE

No one alone.

Every access change is a change set signed by a threshold of admins.



Mai + Ruth approve police break-glass — 2 OF 3 · SIGNED



Dan grants himself the street — 412 QUORUM NOT MET

TIDE IN THE CODE

- TideCloak BYOiD
- DPoP ES256
- doEncrypt / doDecrypt
- 6 tag roles
- AdminAPI change sets
- tide_doctor PASS



"Emergent Authority, made human."

One survivor. Seven parties. No one in charge of the key.

06 WHY THIS MATTERS TO TIDE

Every Tide primitive, in one story anyone understands in a minute.

A reference use case

Shows what "no one holds the key" means for a real person, not a server.

FABRIC · E2EE · BYOID

A public-sector door

Schools, health, courts and police: buyers who must now prove insiders cannot misuse data.

GOVERNMENT · HEALTH · JUSTICE

A governance showcase

Break-glass with witnesses is QEA's clearest demo: two of three sign, one view, revoked.

QEA · THRESHOLD SIGNATURES

Real product feedback

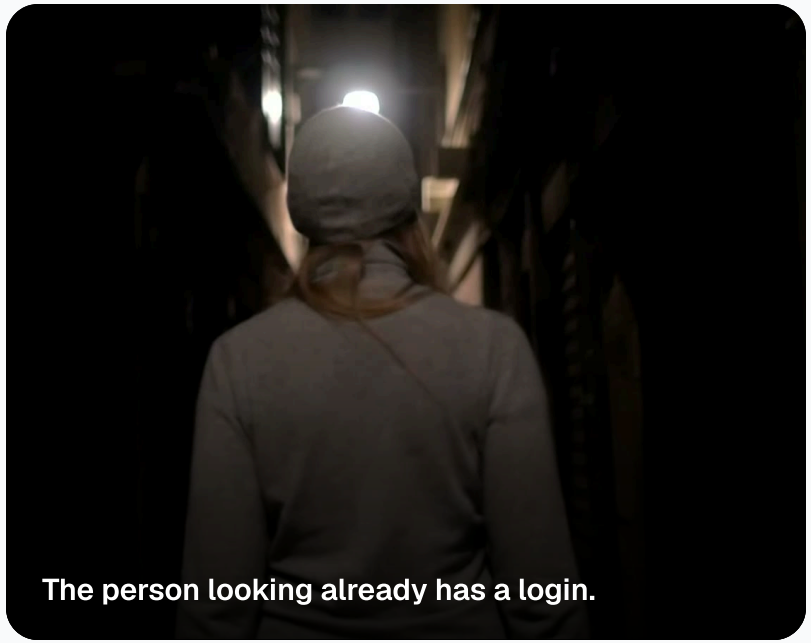
Cross-user decryption, one-time views, cross-org quorums, enrolment health: found by building.

FORSETI · ENROLMENT · SDK

07 THREAT MODEL

What each adversary actually gets.

Unfound is designed for the people who already have access: staff, admins, and us. Each row is enforced by Tide, not by a policy document.

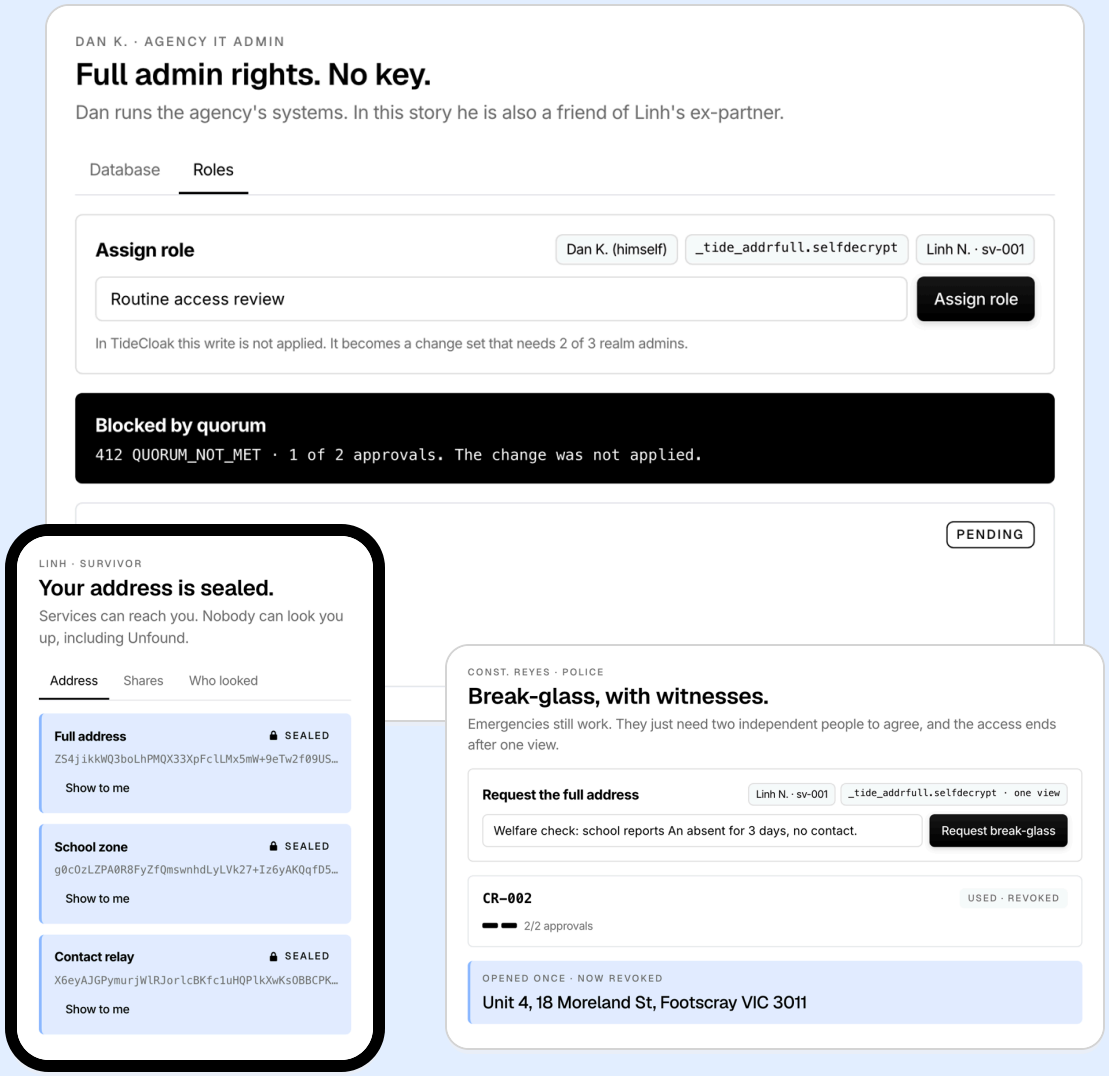


WHO TRIES	WHAT THEY GET	
A curious or bribed staff member	Only the part their role allows; the attempt lands in the survivor's log	DENIED
A realm admin granting themselves access	A pending change set, then 412 QUORUM_NOT_MET	PENDING
Whoever runs Unfound's servers	Ciphertext. No key to steal or subpoena	CIPHERTEXT
A compromised TideCloak server or one ORK	No usable key: keys exist only as fragments across nodes	FRAGMENT
Someone holding a stolen token	Nothing from another device: the token is DPoP-bound	REJECTED
Coercion of the survivor's own device	Out of scope for crypto. Nothing shown by default; quick exit and caseworker recovery on the roadmap	MITIGATED

A working prototype, end to end.

7 personas on iPhone, MacBook and iPad	**68s** auto-played story, 8 scenes	**15** Tide operations per run
0 plaintext addresses stored	**6** `_tide_addr*` roles over 3 tags	**2/3** quorum for any access
3 ✓ tide_doctor checks PASS	**1:55** film, rendered from the app	**0** console errors end to end

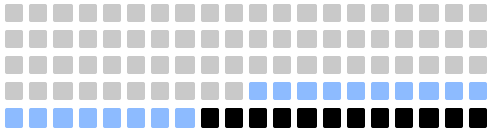
Live path wired: @tidecloak/react 0.14.33, DPoP, doEncrypt/doDecrypt, AdminAPI change sets. First enrolment failed on 22 Oct 2026 when Tide's ORK network answered 3–7 of 20; the public demo runs on a mock fabric with the same rules.



Tide supplies the hard part. The rest is ordinary engineering, and it is funded.

MARKET · AU SCHOOLS

9,653

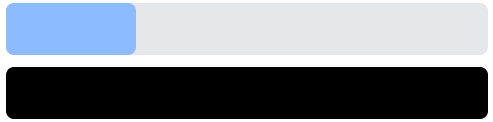


70% government · 18% Catholic · 12% independent

At a hypothetical **A$49 per school per month**: about **A$5.7M a year** from one segment. [9]

FUNDING · AU

A$925M



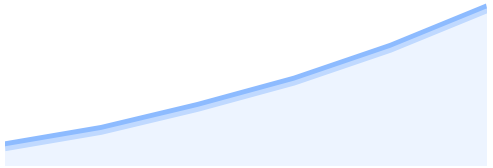
Leaving Violence · A$925M

National Plan · A$3.4B+

Five-year program for people leaving violence, a pathway for funded delivery partners. [10]

PROVEN DEMAND · US

6,212

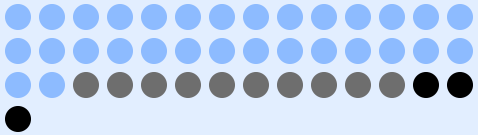


~12% annual growth · over half are school-aged children

Active participants in California's address program alone, run by hand. [11]

PILOT · TIDE FREE TIER

100

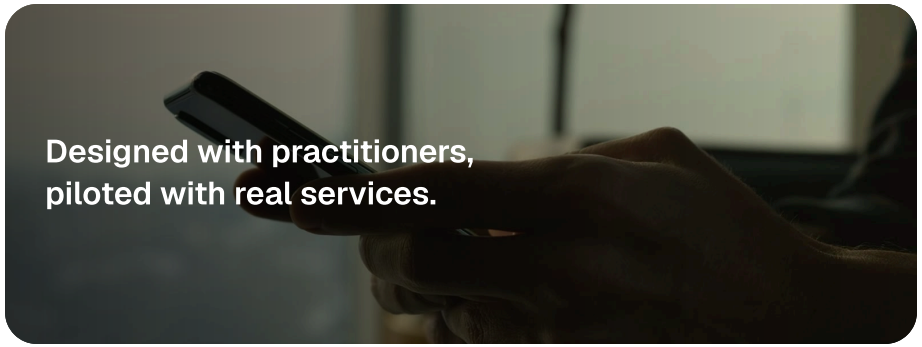


30 survivors · 10 staff · 3 approvers = 43

TideCloak's free tier covers the whole first pilot. [12]

[9] ACARA / ABS Schools 2024; price is a hypothesis to test. [10] DSS Ministers, 2024–25 Budget. [11] CA Secretary of State to NASS, 2023. [12] tide.org/pricing. Accessed 26 Oct 2026.

From prototype to a supervised pilot in 2027.



Nov–Dec 2026 · PHASE 0 — **Jan–Mar 2027** · PHASE 1 — **Apr–Sep 2027** · PHASE 2 — **2028** · PHASES 3–4

Live on Tide

Hosted realm in Australia. Cross-user decryption and one-time views confirmed with Tide.

EXIT CRITERIA

Full story on live Tide; tide_doctor all PASS

Design partners

One family-violence service, one school, one GP clinic. First workflow: enrolling a survivor's child.

EXIT CRITERIA

Signed pilot, PIA, pen test passed

Supervised pilot

20–30 survivors, 3 services, 3 independent approvers.

EXIT CRITERIA

0 plaintext · 100% of attempts visible · onboarding < 15 min

Integrate and scale

School and GP connectors, court print relay, Forseti one-time access. Multi-state AU, Viet Nam pilot. Research: children's own data.

EXIT CRITERIA

Funded contracts in two states

Help us take Unfound from prototype to pilot.

01 **A hosted realm**
TideCloak in an Australian region, plus the Launch Program.

02 **Enrolment clarity**
What caused the ORK failures on 22 Oct, and which health signals to watch.

03 **Two introductions**
A family-violence service and a public agency open to a pilot.

04 **A reference build**
Unfound as a Tide case study once the pilot is live.

FOUR TECHNICAL QUESTIONS

Q1 Can data sealed by one user on tag X be opened by another user who holds `_tide_X.selfdecrypt`?

Q2 Can a Forseti contract enforce “decrypt once, then expire” for break-glass?

Q3 Can one realm's quorum include admins from different organisations?

Q4 Can change-set history be exported per record for a survivor-facing log?

Be reached, not found.

Survivors stop choosing between help and hiding. Agencies can prove that no single person, including their own admins, can reveal where someone lives.

Demo, film, code and brief
github.com/BennedictQuanTon/Unfound-Be-reached-Not-found

built with **TideCloak**

Footage stills: Mixkit (free licence). People: Notionists by Zoish (CC0) via DiceBear. Icons: Lucide (ISC). All records are synthetic.